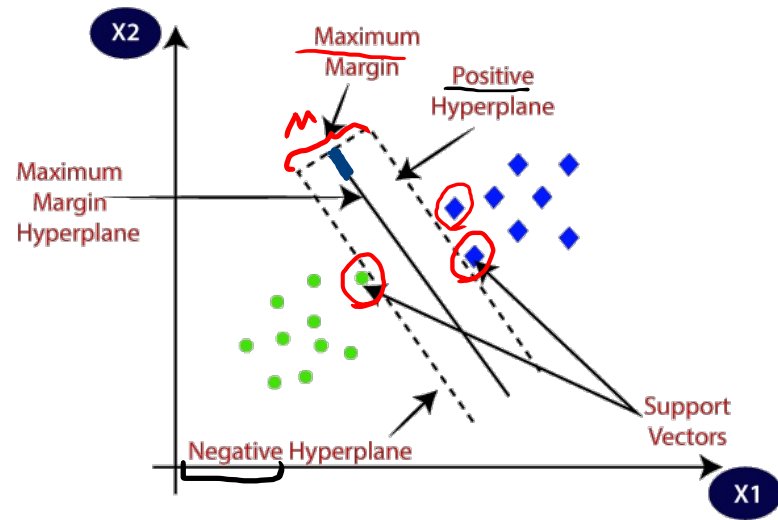
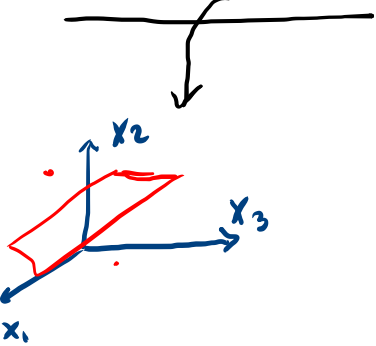


Sum / 2

Hyperplane



(A)

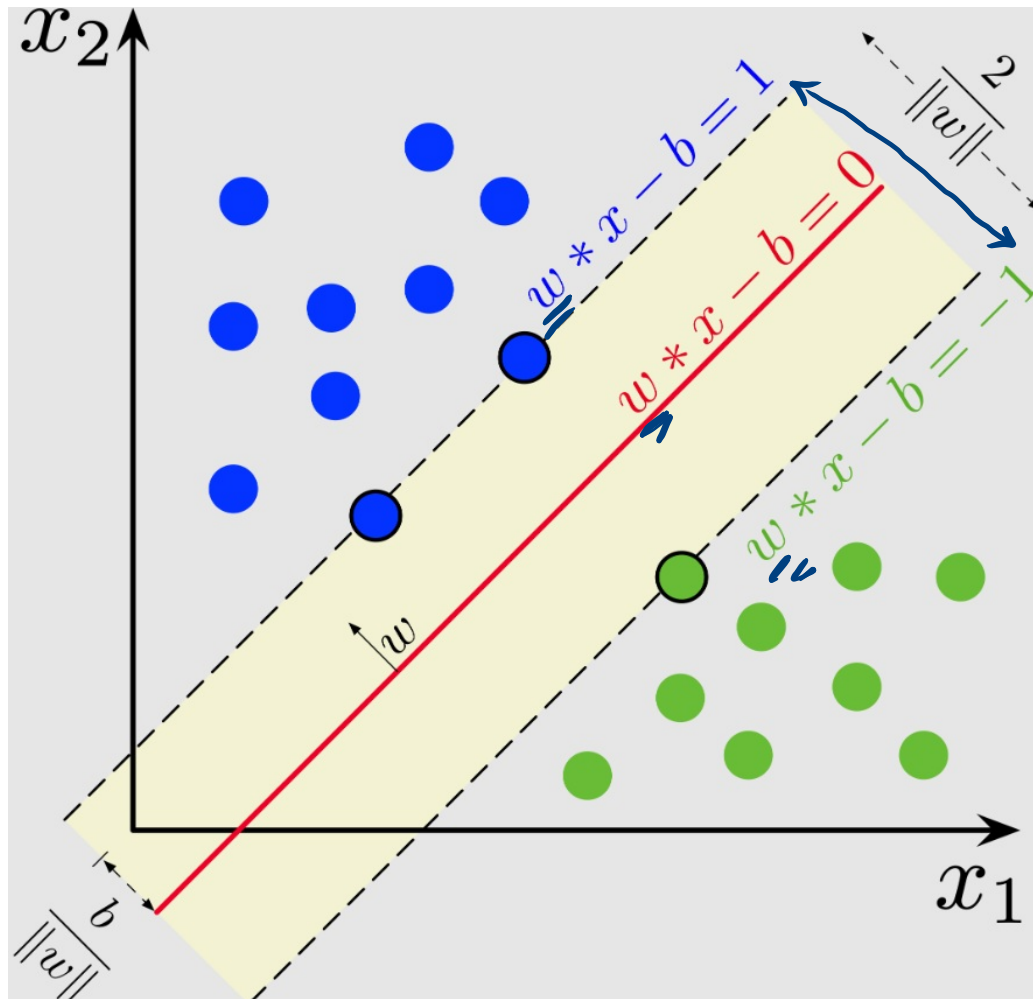
$$\text{Margin} = \frac{2}{\|w\|}$$

Sum $\rightarrow$ Max (margin)

$\downarrow$ min (w)

(B)

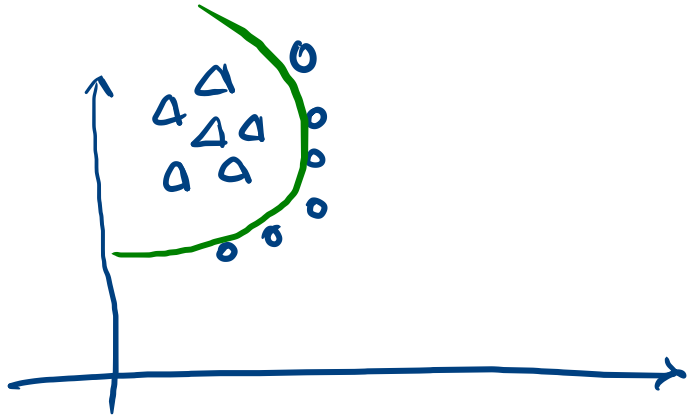
$$y(wx + b) \geq 1$$



Dual form

$$\underbrace{\max}_{\alpha} \left(\sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j \gamma_i \gamma_j \underbrace{x_i^T x_j}_{\alpha} \right)$$

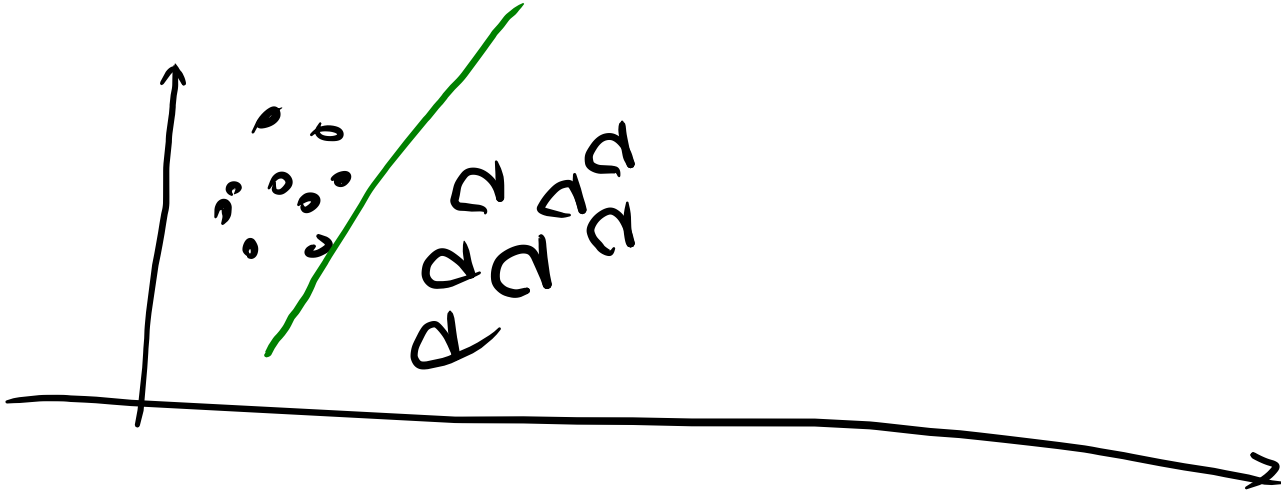
$$\begin{cases} 0 \leq \alpha_i \leq C \\ \sum_i \alpha_i y_i = 0 \end{cases}$$



Kernel Trick

Kernel

Kernel $\rightarrow$ Linear Kernel : $K(x_i, x_j) = \underline{x_i^T x_j}$



$$\underbrace{\text{Dual form}}_{\alpha} \quad \underbrace{\max}_{\alpha} \left(\sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j \gamma_i \gamma_j \underbrace{\left[\underbrace{x_i^T x_j}_{K(x_i, x_j)} \right]} \right)$$

$$\begin{cases} 0 \leq \alpha_i \leq C \\ \sum_i \alpha_i y_i = 0 \end{cases}$$

polynomial kernel

$$K(X_i, X_j) = (X_i X_j^T)^d + r$$

Annotations:

- d (degree) $\rightarrow 3, 4, 5, \dots$
- r (constant) $\rightarrow$ coef 0
- $X_i X_j^T$ (inner product) $\rightarrow$ scaling

RBF kernel

$$k(x, y) = e^{(-\gamma \|x - y\|^2)}$$

HP

← "overfit" فتر

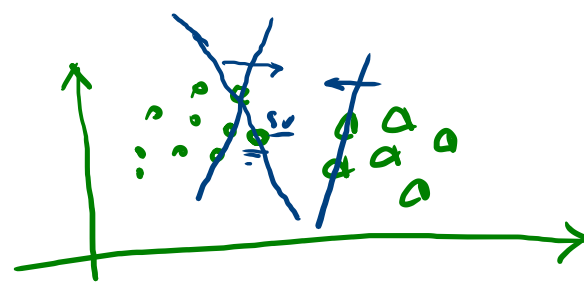
»

← "underfitting" فتر

«

Sum

→ hard margin —



→ soft margin —

